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Profiling Noisy Social Media Data for Sentiment Applications: A Visual and Analytical Framework

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***Abstract* – This article presents a practical approach for analyzing noisy social media data in sentiment analysis. It focuses on identifying and mitigating the impact of elements such as emojis, repeated characters, hashtags, links, and informal expressions, which often distort results. A simple lexicon-based method is applied to assess sentiment both before and after data cleansing. The objective is to illustrate how data quality influences sentiment interpretation, supported by clear examples and interactive dashboards. Results confirm that noisy input affects sentiment scores and can lead to misleading conclusions. The study outlines a structured process for profiling and cleaning textual data, demonstrating how business intelligence tools assist in monitoring and improving data quality. The proposed approach is applicable in practice and enhances the reliability and transparency of sentiment analysis. It also provides a foundation for future advancements in data preprocessing within natural language processing tasks.**

***Keywords* – Data quality, data preprocessing, noisy data, business intelligence sentiment analysis.**

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1. Introduction

Digital technologies are an integral part of people's daily activities, with each action leaving a digital footprint. This has led to an increasing volume of data generated daily across various platforms, particularly in social media. Many companies utilize user-generated opinions to gain insights into trends and attitudes toward specific products and services. The quality of incoming data is a fundamental factor in refining the accuracy of analytical results. The precision and reliability of analyses in the field of Natural Language Processing (NLP), and more specifically, Sentiment Analysis, depend on high-quality textual data to extract meaningful insights regarding user opinions and attitudes. Even minimal inaccuracies in the data, such as spelling and punctuation errors, abbreviations, or misused examples, can significantly affect results [1]. This highlights the necessity of effective preprocessing techniques to minimize the impact of noisy data and enhance analytical performance [2].

Despite advancements in NLP tools, the challenges posed by low-quality input data remain a major concern for researchers and practitioners. Noisy data including typographical errors, auto-generated content, emojis, hashtags, and informal language can lead to misclassification and biased results [3]. This issue is particularly relevant in the context of social media, where short, unstructured, and spontaneous texts dominate. Ensuring higher accuracy in sentiment analysis requires not only technical preprocessing but also early-stage data profiling and evaluation of potential noise.

This study focuses on profiling noisy social media data and visually examining its impact on sentiment analysis outcomes. Instead of relying on advanced models, the approach emphasizes lightweight, lexicon-based techniques that are more interpretable and easier to apply in real-time settings.

Through a visual and analytical framework using tools such as Power BI and Alteryx, the goal is to demonstrate how preprocessing and data quality assessment influence the consistency and interpretability of sentiment signals.

The paper proposes a practical workflow for identifying and mitigating textual noise before sentiment classification.

The findings contribute to answering the question of how much data quality affects the performance of sentiment analysis in noisy digital environments. It is hypothesized that improved input quality leads to more distinct polarity distributions, a lower proportion of neutral classifications, and overall enhanced interpretability of sentiment scores. Additionally, the study outlines a framework for modular data quality profiling that can be integrated into NLP pipelines as a preparatory layer.

2. Literature Review

To provide context for the analysis, the following section reviews key concepts related to data quality and sentiment analysis, highlighting how they intersect in the processing of user-generated content. Different authors [5], [6] emphasize that the concept of data qua lity is based on a complex set of indicators related to the characteristics of the data. According to Wang and Strong [7], the key indicators defining data quality can be grouped into empirical, theoretical, and intuitive categories. In the context of sentiment analysis, data accuracy is crucial, as even minor inaccuracies can lead to significant deviations in the results [4]. Some of the fundamental principles that justify the importance of data quality have also been documented by Jack Olson [8]. He argues that applying techniques to improve data quality reduces systematic errors, enhances business process efficiency, and lowers financial costs for companies. Furthermore, Olson states that poor data are often hidden and require a specially developed strategy for identification. Such data can lead to difficulties in implementing corrections, loss of customers, missed opportunities, and incorrect decision-making.

Accurate data is considered an asset for a company, one that cannot be ensured through standard means.

Achieving high data quality requires careful design of the information system, continuous monitoring of data recording processes in databases, and supervision of processes related to modifying existing data. Olson expands his thesis by providing a perspective on future technological development. He suggests that to ensure high-quality data, companies should develop a tailored data cleansing module that meets the specific requirements of incoming data streams. According to Chomsky's theory [9], the processing of textual data involves stages of morphological, syntactic, and semantic processing. In sentiment analysis, these stages play a crucial role in recognizing emotions in the text, as different linguistic structures can alter the meaning of sentences. Although this theory was developed in 1957, it remains fundamental and foundational for research in linguistics and text data processing. While it has evolved in the context of new technologies and artificial intelligence methods, the core principles of morphological and syntactic processing of language continue to be regarded as essential.

A study in the field of data analysis [10] presents data profiling as a technique for collecting and analyzing metadata to gain a clearer understanding of the current state and characteristics of the data. In another study, Olson [8] describes data profiling as an analytical technique for uncovering the structure, content, and quality of stored data. The outputs of this process include reliable metadata and data quality indicators, which are instrumental in developing specialized procedures to improve data quality within a given system.

In summary, profiling can be viewed as a comprehensive, multi-component process involving the examination and analysis of dependencies within existing data by applying data testing techniques. As a result of this process, based on predefined rules, the data is divided into two groups: valid and invalid. Invalid data is categorized according to the nature of the errors it contains and is separated from the main data flow for independent review and classification.

In sentiment analysis, which relies on natural language processing, the focus is on the automatic identification and categorization of emotions and feelings expressed through written text, as illustrated in Figure 1.

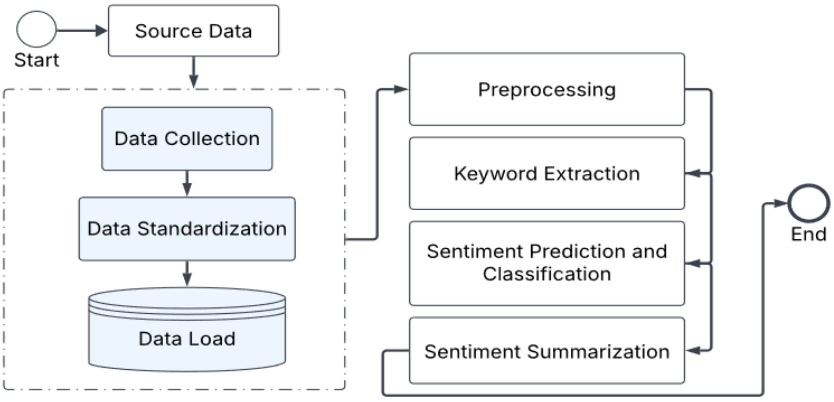

Figure 1. A general framework of sentiment analysis process [15]

Research shows that noise in textual data significantly reduces the accuracy of sentiment analysis algorithms. For example, Baccianella and authors [3] found that statistical methods for analyzing text length and error levels can identify potential data issues and eliminate them during preprocessing. The sentiment analysis process consists of several key steps, including data preprocessing, feature extraction, and classification. During the preprocessing stage, basic transformations are applied to the input data to remove irrelevant information such as stop words, special characters, and numbers [11]. These transformations are specifically tailored to the analysis of user opinions and sentiment. This stage also involves converting textual data into a structured sequence of words using various techniques.

Existing research [12], in the field of NLP highlights the importance of data quality in improving analytical outcomes. According to the study, noise in textual data reduces the effectiveness of trained models and leads to deviations in predictions. At the same time, emphasizes that standard preprocessing techniques, such as stop word removal and stemming, are insufficient for eliminating all sources of noise. Other NLP studies, such as those by Gosh et al. [13], indicate that automated spelling correction systems can reduce noise in texts and enhance the accuracy of classification models. From the perspective of text data processing and transformation, no universal algorithm has been developed, as text data is highly specific in both structure and formation [14].

Scientific research based on studies of various types of textual data provides recommendations for improving methods and techniques to achieve greater precision in processing [4], [15], [16], [17]. In recent studies on sentiment analysis, data quality plays a foundational role in achieving accurate and reliable results.

Shad and authors emphasize thatcomparisons between different machine learning algorithms in the context of sentiment analysis show that models with high-quality data yield significantly better results [18]. Other authors confirm that, despite the power of machine learning, it heavily depends on the quality of textual data, with poor-quality data leading to serious inaccuracies in sentiment classification [19].

Additionally, Dakwah and authors highlights that applying sentiment analysis for political recommendations requires utmost attention to data quality, as inaccurate or noisy data can lead to erroneous interpretations [20]. Kumar and Garg also note that when analyzing both handwritten and electronic text documents, data quality is crucial for the accuracy of sentiment analysis algorithms [21]. Furthermore, Bhardwaj and Girdhar demonstrate that accurate sentiment analysis of social media texts requires high-quality data to minimize noise and ensure correct emotion classification [22]. These sources clearly show that, despite advancements in algorithms, data quality remains a key factor for the success of sentiment analysis. The framework adopted in this study operationalizes these dimensions using tools such as Alteryx and Power BI, allowing the visual profiling, cleansing, and analysis of user-generated content prior to sentiment interpretation through VADER.

3. Data Profiling and Quality Enhancement

The analysis was conducted on a dataset of 1,000 short text entries related to the COVID-19 pandemic, reflecting the informal and noisy nature of social media content. Sentiment scores were assigned using the VADER tool, integrated into Alteryx workflows for streamlined processing and traceable transformations. No gold standard labels were applied, as the focus was on internal consistency and the impact of data cleansing on polarity shifts rather than external benchmarking.

Due to the diverse nature of the data, it is necessary to apply an individualized approach during the initial exploration and implement predefined rules, including checks for missing values, duplicate values, incorrect values, grouping by similarity, summarization of minimum and average values, and measurement of data length in characters, consistency checks, and more. The study proposes the development of a separate module to verify and, if needed, enhance the quality of the incoming data stream. The proposed conceptual framework is visually presented in Figure 2.

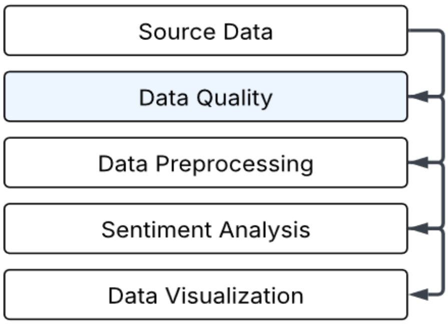

Figure 2. Conceptual model of sentiment analysis process with data quality module
Source: author's own research

The analysis of data against predefined rules (Value Rule Analysis) is based on statistical methods that calculate key metrics such as minimum value, maximum value, domains, frequency distribution, standard deviation, and more [10]. These calculations are applied to a subset of the data, with results that can be presented as a report, a dashboard, or stored in a file. This type of analysis helps identify anomalies in the data, such as deviations from standard, minimum, or maximum values. The results obtained from profiling the incoming data are transformed into rules that aid in improving data quality. By its nature, data quality is of fundamental importance for the proper functioning of business processes and is considered an indicator of whether the data meets the requirements for its intended use in a business system [8].

Therefore, data quality largely depends on its expected application rather than being evaluated as an independent structure.

The key indicators for measuring data quality, as proposed by Olson [8] and supported by other research studies [5], include the following:

(a) data consistency - relates to the validity of the data and verification of its structure and format. It reflects the relationships between real-world objects;

(b) data completeness - represents the extent to which the data contains all the necessary information about the objects. This indicator measures the amount of missing data and can be calculated using the following formula:

$$\text{Completeness rating } (\boldsymbol{CR}) = 1 - \left(\frac{total\ number\ of\ data\ objects\ with\ missing\ data}{data\ objects\ total\ number}\right)$$

Completeness rating – indicator which measures that all related data objects are presented.

(c) data accuracy - represents the permissible values according to the established requirements. This metric indicates the level of accuracy by comparing the data in the system with its corresponding real-world object; (d) data timeliness - determines the relevance of the data based on two aspects: the time elapsed since the last data update and the frequency of data modifications.

A technique is applied to separate records based on their timestamp in the system. The outlined characteristics of data profiling and the associated tasks are visually represented in Figure 3. During the data profiling process, a parallel analysis of metadata related to the incoming data is also performed. The entire incoming data stream is checked for missing values, duplicate entries, incorrect data, potential data patterns, and categorized based on its overall structure. Additionally, the minimum, average, and maximum values are identified, along with the length of each user opinion measured in characters.

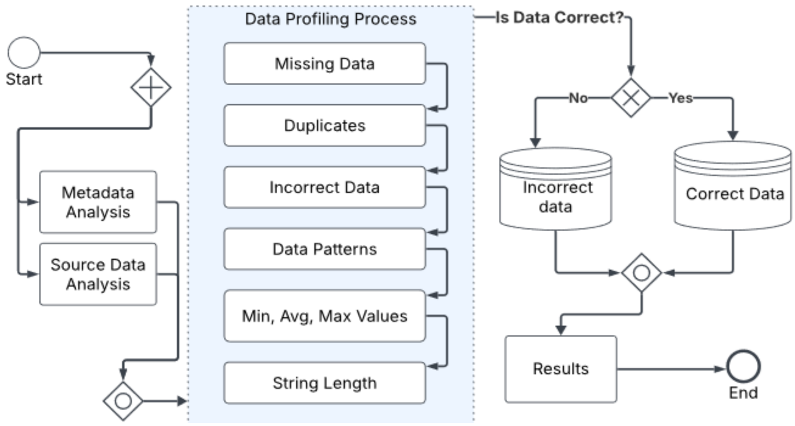

Figure 3. Conceptual framework for data profiling module
Source: author's own research

According to predefined data validity rules, the data is classified as either correct or incorrect. Each group is stored separately to ensure accessibility for subsequent analyses. The examined indicators characterize the properties of the data as measured during the preparation process before loading it into a database. A summary of the characteristics that define data quality and their application in natural language analysis is presented in Table 1.

Table 1. Data quality indicators in Sentiment analysis

Indicator	Description	Example
Consistency	Are the data in the same format?	All dates are in the same format, e.g., DD/MM/YY
Completeness	Is a part of the data missing?	The user opinion is recorded in full without edits or abbreviations.
Accuracy	Are there any errors in the data?	Does the expressed opinion correspond to the user's profile?
Timeliness	Are the data up to date?	Are the user's opinion recorded from the time of publication?

Source: author's own research

Other researchers [2] support this argument in their publication, but add the third dimension related to the human factor in data management within an information system. The author emphasizes that data processing specialists are responsible for the proper storage of data definitions, business rules, domains, and their updates.

From the perspective of conducting a precise analysis of user opinions and capturing sentiment, it is crucial to establish a dedicated data quality enhancement module within the overall data analysis process.

This module should be designed based on the nature of the incoming data and the results of the initial data profiling. The conceptual model, illustrating the sequence of actions and associated objects within the overall data analysis process, is visually represented using the Business Process Model and Notation standard in Figure 4.

The data analyzed is then directed to the data quality improvement module, where custom-developed rules are applied to refine the data, following established techniques. Data cleansing techniques are among the traditional methods for improving data quality. They are applied after data profiling to values categorized as unacceptable for a given dataset [23].

Data cleansing aims to resolve existing issues, such as duplicate values in columns expected to contain unique entries, inconsistencies in functional dependencies, missing values, sequencing errors, and other structural inconsistencies [10]. Inaccuracies in the data are typically corrected by modifying the erroneous values either by replacing them with predefined values, removing them, or generating new ones. Different approaches to data cleansing and normalization exist, but the preferred method is the one that minimizes alterations while effectively eliminating the identified inconsistencies. In his book, Loshin [24] highlights data cleansing as one of the most crucial techniques for correcting "bad" data and bringing it into a standardized form.

On the left side of the figure, the incoming data is displayed, which is extracted from the Kaggle platform. These data are openly accessible and do not violate confidentiality or personal data protection regulations. Their practical impact is expected to be evaluated at a later stage during the development of a sentiment analysis prototype, which will be presented as an experiment.

The data quality enhancement process begins with the intake of data for processing. The next step involves profiling the incoming data to comprehensively assess its condition (Figure 4).

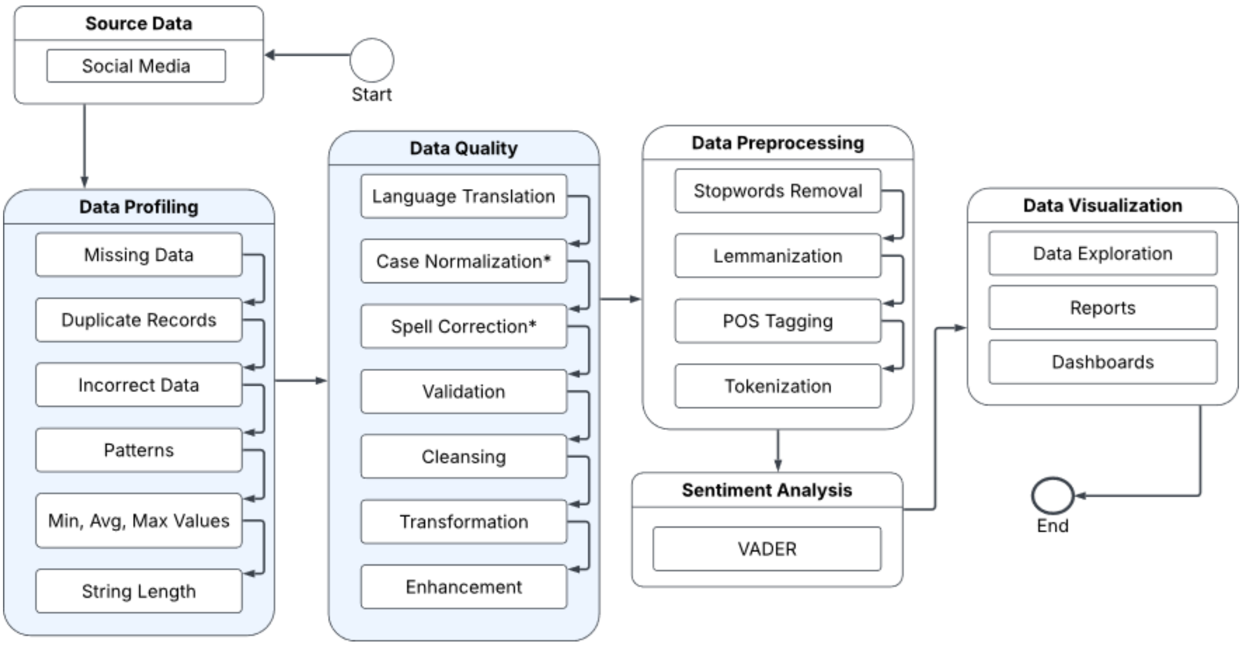

Figure 4. Integrated workflow for social media sentiment analysis based on data profiling and quality improvement
Source: author's own research

Other researchers expand the concept by likening it to a process for identifying and correcting erroneous data, which serves the following purposes: (a) re-moving invalid characters within a dataset; (b) separating multiple pieces of in-formation within a single entry; (c) standardizing unstructured data; (d) enriching incomplete data; and (e) transforming existing data to align with specific business requirements [25].

Based on the requirements set for the process and the nature of the incoming data, the last two actions mentioned above data enhancement and data trans-formation can be categorized as two separate groups of operations aimed at improving data quality [26]. Among the activities related to data transformation, we can include Language Translation, Case Normalization, and Spell Correction. In the next stage, a data validation technique is applied to verify whether the values stored in the system comply with the permissible domains. These domains are defined according to business needs and are implemented in the system as a catalog containing only up-to-date and correct values [24].

The development and implementation of validation rules are considered standard techniques for assessing data quality [8]. Due to the high flexibility of validation algorithms, a rule can be extended to incorporate multiple statistical formulas and integration with external sources. Once the data quality improvement rules have been applied, the next step (Figure 4) involves executing sentiment analysis specific transformations, which will be performed on the cleansed and standardized data.

The first step in the preprocessing part is stop words removal. This involves filtering out frequently occurring words in each language (e.g., "and" "but" and "the") that carry little semantic weight. Removing stop words is a standard step in Natural Language Processing (NLP) since these words typically do not contribute to the understanding of a text's meaning and may hinder analysis [27]. This step reduces data size and enhances the efficiency of machine learning algorithms.

Next, lemmatization is applied, where words are reduced to their base (lemma) form based on their contextual meaning. For example, the words "running", "ran", and "runs" are all reduced to "run". Unlike stemming, which removes suffixes using simplified rules and is not the focus of this study, lemmatization relies on linguistic dictionaries to ensure grammatically and semantically correct outputs.

POS tagging (Part-of-Speech tagging) is the process of automatically determining the grammatical category of each word in each text. For example, in the sentence "The cat sleeps on the couch", the POS tagger would label "The cat" as a noun (noun), "sleeps" as a verb (verb). POS tagging supports syntactic analysis, machine translation, and various NLP tasks. Tokenization is applied at the end of this process step. It splits the text into individual components (tokens), which can be words, sentences, or other meaningful units. Tokenization is a fundamental NLP step that facilitates further analysis. After this specialized data preparation phase, the processed text is directed to the sentiment analysis module.

After defining the key steps involved in data preprocessing, including stop word removal, lemmatization, POS tagging, and tokenization, the next stage involves the practical implementation of these transformations through a custom-developed application. This application integrates the outlined processes into an end-to-end data pipeline for sentiment analysis.

To operationalize the described data preparation techniques, the study employed a custom-built application that integrates the outlined processes into a complete sentiment analysis pipeline. This implementation ensures that each preprocessing step from lexical normalization to syntactic annotation is executed consistently and reproducibly. The technical environment supporting this implementation is described below.

The implementation of the proposed approach was carried out using the Alteryx Designer development environment. Alteryx was selected due to its integrated Intelligent Suite for advanced analytics and its flexibility in enabling user-defined configurations without extensive coding. Compared to other data quality platforms such as Ataccama Data Quality Center, Talend, Informatica Data Quality, and Observation, Alteryx offers a user-friendly interface and streamlined integration of data profiling, transformation, and automation tasks. According to Gartner's 2024 report [28], Alteryx is recognized as one of the leading niche players in the category of Data Science and Machine Learning platforms.

As part of the visualization component of the framework, Power BI enabled side-by-side tracking of sentiment polarity and data cleansing impact. Its accessibility, integration with diverse data sources, and native support for DAX measures made it suitable for building a responsive dashboard interface aligned with the analytical goals of the study. Compared to platforms like Tableau and Qlik, Power BI offered a more seamless integration with Microsoft-based workflows and was better suited for rapid prototyping in academic research contexts [29].

Despite the strengths of the conceptual framework, several limitations should be acknowledged. The processes of data profiling and quality enhancement are computationally intensive, particularly when applied to large-scale or heterogeneous datasets. The framework, in its current form, is optimized for moderate-sized, context-specific datasets, such as social media posts related to targeted events or public discussions. This focused scope facilitates the detection of anomalies and the removal of noise but inherently restricts the model's generalizability across broader or more dynamic topics.

Additionally, the effectiveness of individual data cleansing and profiling techniques such as eliminating irrelevant elements or correcting inconsistencies is highly dependent on the data domain. Therefore, the impact of the framework may vary significantly depending on the structure, semantics, and topical orientation of the input data sources.

4. Lexicon-Based Sentiment Analysis and the Impact of Data Profiling

This section examines the impact of data profiling on sentiment detection when using a rule-based, lexicon-oriented approach. Rather than focusing on deep-learning models, the analysis relies on the Valence Aware Dictionary and Sentiment Reasoner (VADER) sentiment lexicon, which assigns polarity scores to words based on their perceived emotional intensity.

The VADER lexicon uses a predefined sentiment polarity scale ranging from -4.0 (extremely negative) to +4.0 (extremely positive). Each word in the lexicon is scored based on human ratings, taking into account factors such as intensity, context, and usage in informal settings such as social media [17]. Neutral terms are generally assigned a score close to 0.0. Sentiment analysis using this approach calculates the aggregate polarity of a text by summing or averaging the scores of the individual words it contains (see Table 2).

Table 2. Example of sentiment-bearing terms from the VADER lexicon

Lexicon Term	Polarity Score
excellent	+2.0
terrible	-2.5
safe	+1.5

After data cleansing, previously unrecognized or distorted expressions (e.g., slang, informal spelling) are normalized, enabling them to match entries in the sentiment lexicon. This increases the accuracy and coverage of the sentiment analysis.

For example, in the original noisy dataset, tweets like "Vaccs wrk!! finallyyyy feelin safe!" may not have matched any known lexicon items. After cleansing ("Vaccines work. Finally feeling safe"), the normalized version allows better semantic alignment and scoring precision. To illustrate the effect of data cleansing, selected user-generated texts were analyzed in their original and processed forms and structured in Table 3. The applied transformations included removal of duplicate entries, standardization of formats, correction of common spelling errors, and normalization of letter case and sentence structure.

Table 3. Examples of sentiment correction after data cleansing

Source Text	Output Text	Sentiment Polarity
"I h8 this app!! ughhh 😡 "	"I hate this app."	Negative
"luv it soooo much 😍 😍 😍 "	"Love it so much."	Positive
"idk what2say meh"	"I do not know what to say."	Neutral
"finallyyy!!! safe & sound"	"Finally! Safe and sound."	Negative

Source: author's own research

The cleaned versions exhibit significantly improved readability and semantic clarity. As a result, sentiment scoring especially when using lexicon-based approaches such as VADER becomes more precise and reliable. The sentiment distribution before and after cleansing was visualized using Power BI. A comparative word frequency analysis and polarity histogram revealed that noisy texts often led to skewed sentiment values due to non-standard spelling, emojis, or fragmented expressions. After profiling, the sentiment distribution became more balanced, with a notable reduction in misclassified neutral entries and a clearer separation between positive and negative cases. Several notable improvements were identified after applying data profiling and cleansing techniques to the initial dataset. These enhancements affected both the interpretability of the data and the performance of the sentiment analysis models:

Reduction of ambiguity: The removal of noise such as duplicated records, excessive punctuation, and informal expressions enabled sentiment models to perform more confident and precise classifications. Cleansed texts exhibited a clearer semantic structure, which reduced the likelihood of neutral or incorrect categorization due to unclear language.

Improved lexical coverage: The normalization of abbreviations, misspellings, and informal expressions significantly increased the alignment between user-generated content and predefined sentiment lexicons. As a result, previously unrecognized phrases were successfully interpreted, enhancing the overall analytical scope and granularity of the sentiment detection process.

Enhanced contextual representation: Standardized syntactic structures and corrected word forms facilitated more accurate mapping between textual inputs and the sentiment scoring framework.

This was particularly beneficial for rule-based methods, where lexical integrity plays a crucial role in capturing nuanced attitudes and emotional expressions.

These observations underscore the critical role of data quality in natural language analysis workflows. Even when the underlying model architecture remains unchanged, the input quality determines the interpretability, reliability, and richness of the extracted insights. In this regard, data profiling and cleansing should not be viewed as auxiliary steps, but as integral components of the sentiment analysis pipeline capable of directly influencing the accuracy and practical value of the results.

5. Research Results

Following the methodological steps outlined in the previous section, this part presents the results of the experimental implementation, focused on profiling, cleansing, and analyzing noisy Twitter data related to the COVID-19 pandemic.

The main objective was not to optimize a sentiment classification model, but rather to observe how basic data profiling and quality improvement techniques affect the clarity and interpretability of the sentiment analysis outcomes.

To achieve this, a three-step workflow was designed: initial data profiling, quality enhancement through standardized transformations, and sentiment analysis using the VADER tool. The outputs were then visualized and compared using a dynamic Power BI dashboard. Each stage is described in more detail below.

As previously described, the initial step involved an exploratory profiling of 1,000 user-generated tweets collected during the COVID-19 pandemic.

These tweets were selected to reflect a realistic mix of opinions, expressions, and language inconsistencies often present in social media content.

Using Alteryx Designer, a profiling workflow was developed to scan the dataset for typical quality issues, such as missing values, inconsistent casing, excessive punctuation, elongated words, emojis, and duplicate entries. Figure 5 presents a snapshot of the profiling process, highlighting key metrics such as word count distribution, frequency of non-standard characters, and token patterns.

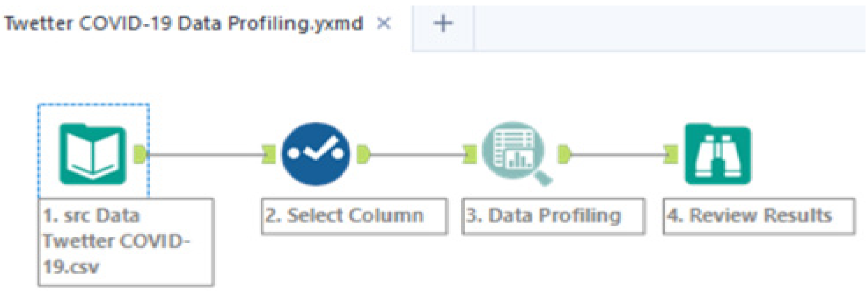

Figure 5. Profiling of user-generated content using Alteryx Designer
Source: author's own research

This stage allowed the identification of patterns that could interfere with the lexical interpretation of sentiment-bearing terms. It also provided a quantitative baseline for measuring the impact of subsequent cleansing steps.

After profiling, a tailored cleansing procedure was applied, focusing on minimal yet effective transformations. These included: normalization of spelling and letter case, removal of non-alphabetic noise (e.g., excessive punctuation, repeated characters), standardization of emojis into textual descriptions, and elimination of redundant or empty entries.

The transformation logic was implemented as a sequence of tools in Alteryx, ensuring replicability and traceability. Figure 6 shows the structure of the cleansing workflow. This stage aimed to improve syntactic consistency and to optimize the input for lexicon-based sentiment scoring. Instead of eliminating informal language, the goal was to structure it for better interpretability.

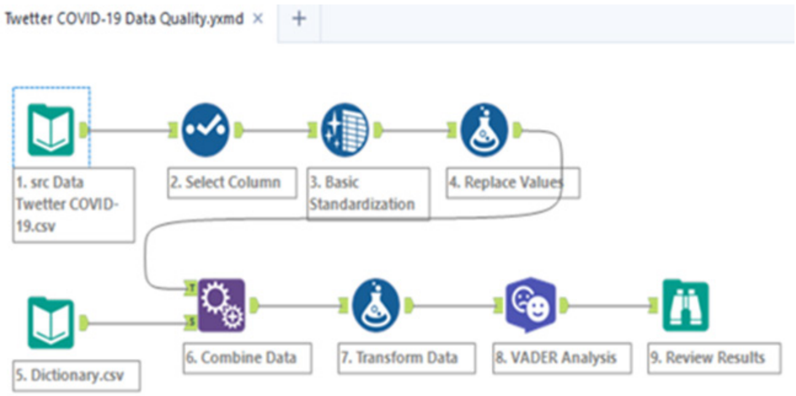

Figure 6. Data quality enhancement workflow for sentiment preprocessing
Source: author's own visualization

This workflow was not designed to radically alter the content, but rather to preserve the original semantics while improving formal characteristics. The guiding principle was to enable better alignment between user-generated expressions and lexicon entries used in sentiment detection.

As part of the validation process, intermediate outputs were inspected to ensure that sentiment-bearing words were not unintentionally removed or distorted.

Particular attention was given to preserving:

- sentiment modifiers (e.g., very, not);
- informal but emotionally loaded terms (e.g., meh, yay);
- and standardized forms of expressive tokens (e.g., emojis → ":heart:", ":angry_face:").

This balance between cleaning and retaining context played a critical role in maintaining the expressive richness of social media content while enhancing its computational interpretability.

Following the data profiling and cleansing processes, a comprehensive Power BI dashboard was developed to evaluate the outcomes of the sentiment analysis and provide visual control over key data quality metrics. The dashboard is presented in Figure 7. It integrates outputs from Alteryx workflows and allows dynamic exploration of sentiment shifts, term frequencies, and classification distributions across the dataset.

The core purpose of this dashboard is twofold: first, to verify the effectiveness of the cleansing pipeline by comparing sentiment scores before and after transformation; and second, to provide stakeholders with a clear and visual understanding of how noise and structural inconsistencies in the data influence sentiment interpretation.

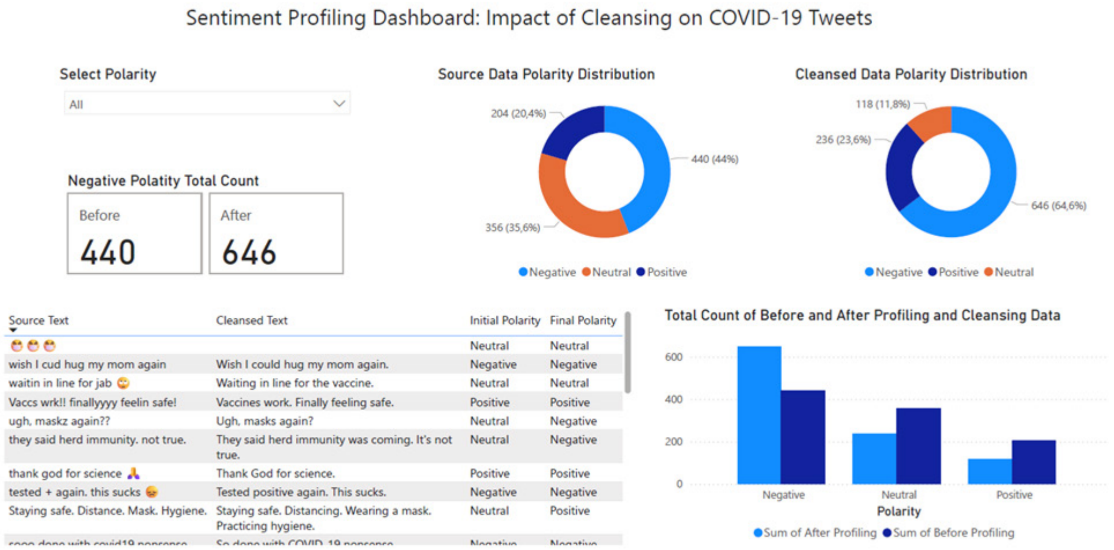

Figure 7. Interactive dashboard created in Power BI, presentingsing sentiment polarity distribution, term frequency insights, and textual examples
Source: author's own research

The final stage of the research involved the development of an interactive Power BI dashboard that summarizes and validates the impact of data cleansing on sentiment classification outcomes. The dashboard brings together processed outputs from Alteryx and enables a direct visual comparison between sentiment scores assigned before and after profiling.

The core purpose of this dashboard is twofold: first, to verify the effectiveness of the cleansing pipeline by comparing sentiment scores before and after transformation; and second, to provide stakeholders with a clear and visual understanding of how noise and structural inconsistencies in the data influence sentiment interpretation.

The dashboard is organized into several key components. A clustered bar chart illustrates the shifts in sentiment distribution across the three categories (positive, negative, and neutral), based on a sample of 1000 tweets. This visualization clearly reflects the reduction of neutral misclassifications and the sharper delineation between positive and negative classes after cleansing. In parallel, numeric indicators provide an at-a-glance view of the percentage changes across sentiment categories, quantifying the gain in classification precision.

In addition, a table presents selected tweet samples alongside their original and cleansed versions, allowing users to explore specific text-level changes and their impact on sentiment polarity. This contextual layer enhances transparency and supports qualitative evaluation of the transformation pipeline.

The dashboard thus serves as a bridge between technical preprocessing and strategic insight, enabling non-technical stakeholders to appreciate the tangible effects of data quality enhancement in a sentiment analysis workflow.

6. Conclusion and Future Work

This paper presented a practical framework for profiling and cleansing noisy textual data in sentiment analysis applications. Focusing on user-generated content from social media, the research demonstrated how systematic preprocessing through data profiling, standardization, and transformation enhances both the interpretability and accuracy of lexicon-based sentiment models. Using VADER as an accessible, rule-based approach, the study emphasized the importance of input quality in achieving meaningful analytical outcomes.

The integration of Power BI visualizations and Alteryx workflows provided a transparent and reproducible method for assessing data quality and sentiment classification results. The findings confirmed that even modest improvements in data structure and lexical consistency can lead to measurable enhancements in sentiment polarity interpretation. These outcomes are particularly relevant for business, academic, and policy contexts where decisions are informed by social media insights.

The primary contribution of this research is the integration of a data quality assessment workflow within a sentiment analysis pipeline, demonstrating that even simple lexicon-based models benefit significantly from structured preprocessing. Additionally, the proposed framework provides a replicable example of combining low-code tools to enhance interpretability in applied NLP contexts.

The proposed workflow is adaptable to various domains and can serve as a template for further research or operational applications where data quality is a limiting factor in natural language processing tasks.

Future work will focus on expanding the framework to accommodate multilingual data sources and to evaluate the effects of data cleansing across different sentiment analysis techniques, including transformer-based models. Another avenue for development includes incorporating feedback mechanisms and dynamic data validation into the cleansing pipeline, ensuring adaptability in real-time analytical environments. The integration of machine learning methods remains a promising direction, as they offer scalable, automated analysis of complex datasets and support timely decision-making in data-intensive applications [30]. Further refinement of the dashboard's interactivity may also support its application in decision support systems and stakeholder reporting.

By establishing a clear link between data quality and analytical precision, this study contributes to the growing body of research that recognizes preprocessing not as a preliminary step, but as a critical component of natural language processing workflows.

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